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1. Introduction

This paper summarizes the work carried out during the Aerospace Technologies Laboratory. The aim of the laboratory was to analyze and study the advantages and problems of 3D printing using a lunar regolith simulant. Using *Autodesk Fusion®*, a bar with XY orientation and a cylinder with a gyroid lattice structure were designed.

The modeling parameters are shown below.

Flexural bars	X	Y	Z
Size	40 mm	5 mm	3 mm

Cylinder			*Offset (2 base-faces)*			*Gyroid infill*
Dimension	*Diameter*	*Height*	*Thickness*	*Blend distance*	*Solidity*	*Uniform*
Size	8 mm	16 mm	2 mm	1 mm	1.00	2 mm - 60 %

Using the Chitubox® *slicing program*, the process parameters for the print were set. Taking into account a *layer height* of 30 µm, four prints of the XY bar were made, of which three were sintered, and three cylinders, of which two were sintered.

With the aim of better addressing the multiple sections of which this report is composed, the group was divided into several subgroups, each of which addressed a different thematic area in greater detail: materials, process parameters, quality of the components produced and mechanical properties.

Clearly, for each thematic area it was necessary to use different research techniques: in the field of materials, we relied on a comparison of the data provided, with insights related to the specific consequences of the composition of the regolith. As far as the process parameters are concerned, those we considered most critical were highlighted, as a result of the R.A.M.S. analysis. The quality was verified through the analysis of microscope images, using numerous software, including MATLAB® and ImageJ®. Finally, the mechanical properties were tested in the laboratory thanks to specific machinery: the bars were flexurally tested, the cylinders were compressed.

Once the report was completed, all the subgroups gathered actively participated in the drafting of the text of the slides for the oral presentation.

2. R.A.M.S. analysis of the production process

2.1. Materials

The material used for 3D printing is a custom mixture composed of 30% custom resin, obtained from oligomers and photopolymers, and 70% *LHS-1,* a solid mixture of powders created to simulate the behavior of lunar regolith.[1]

To control the production process and better understand the characteristics and properties of the finished product, it is useful to characterize both components. The *LHS-1* solid mixture has a composition derived from samples acquired as part of lunar exploration. It consists mainly of anorthosite (74.4%) and glass-rich basalt (24.7%) particles. Chemically, it is composed of numerous oxides, including SiO_2, Al_2O_3 and CaO. The composition has some effects on the production process: the oxides absorb and diffuse UV rays and interfere with the polymerization process of the resin. The exposure time therefore becomes a critical parameter for the process.

In addition to the chemical composition, the particle size is also a fundamental parameter to establish the physical properties (thermal and mechanical) of the mixture that will influence the results of 3D printing. For this reason, a comparison was made between the actual particle size distribution and that of the manufacturer, which showed a discrepancy with the data. In fact, all the particles in the sample are less than 8,710 µm in size, compared to an expected percentage of less than 12%. A greater quantity of smaller particles translates into an increase in the total specific surface area, an increase in viscosity and a decrease in the polymerization rate. As a result, a greater quantity of resin is used, an increase in the probability of agglomeration and, therefore, of printing defects, as will also be seen in the following points.

2.2. Process parameters

The process parameters set for printing are part of the R.A.M.S. analysis of the project[2]; si report the parameters defined as critical.

- Reliability (**R**. A.M.S.)
  The printing process must work by minimizing any defects in the parts produced in a defined period of time. Working times are, therefore, an essential element of reliability: exposure time is adopted as a critical parameter of the process (indicated as 15 s). In addition, another key concept is the reduction of the possibility of printer failure. By choosing moderate *lift* and *retract* speeds (40 and 80 mm/min respectively), there is less wear on the printer's mechanical arms during the production cycle, with a consequent reduction in the probability of machine failure.

- Availability (R.A.M.S.)
  The 3D printer must be operational at the request of the operator. Excessive *lift* and *retract* distances lead to increased printing times with a consequent reduction in machine availability, the *lifting distance* is chosen as a critical parameter (set at 10 mm). Similarly, the overall rest time between one layer and the next, obtained as a combination of *rest time before lift*, *after lift* and *after retract* (combined they provide 21 s of waiting), increases the possible downtime.

[1] The percentages are intended with respect to the weight

[2] Printer considered: *Elegoo Mars 4 DLP®*

- Maintainability (R.A. M.S.)
  Need to perform simple and targeted interventions on machinery and process. Adequate *lift* and *retract* speeds such as those mentioned above, allow the correct calibration of the z-axis to be maintained during printing, therefore less need for intervention for any realignment.

- Safety (R.A.M. **S**.)
  Minimization of risk for user, operator, machinery and/or component. An incorrect power level leads to damage to the operator, if in excess, or an incomplete polymerization of the resin, if in defect: hence the power of the source (18 W from *the Elegoo Mars 4 DLP®* data sheet) as a critical parameter indicating safety in the work area.

It is important to note that each parameter can influence a different part of the R.A.M.S. analysis depending on its interpretation. Take the *transition layer time* as an example: it guarantees greater reliability thanks to the reduction of any mechanical defects between the base and the body of the part, at the same time it causes an increase in exposure times negatively impacting availability.

3. Quality of the components produced

To evaluate the quality of the components produced, some of their dimensional and defect characteristics were analyzed through a microscopic image analysis. Mechanical resistance tests were also conducted. The evaluation was also carried out with reference to the critical process parameters identified.

3.1. Dimensional tolerances

For component analysis, data was acquired using light and scanning electron microscopes[3], for observations at different scales. Processing was conducted using MATLAB® and ImageJ®, used individually or together, to exploit their specific functionalities and obtain the desired data with maximum precision.

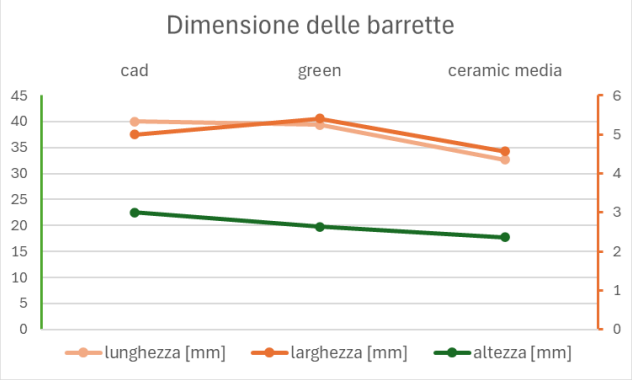
The graphs show the nominal measurements of the bar imposed in the CAD model and the experimental measurements made with digital caliper. The printed dimensions do not correspond to those of the green specimen (not sintered): length and height are smaller than the nominal ones, while the width is greater. Calculating the percentage shrinkage between model and sample, a negative shrinkage (therefore a widening) in the width is noted. Possible reasons for the difference between CAD and green are to be found in the process parameters described in the R.A.M.S. analysis, in particular the rest time between successive layers and the correct alignment of the z-axis.

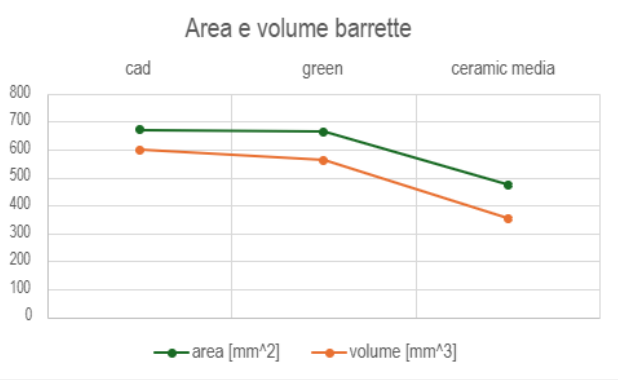
By comparing the data of the green sample and the average of the sintered ones, an estimate of the percentage shrinkage of the sample can be obtained. Before sintering, the resin is removed from the sample through the *debinding* process, in this way the temperature can be increased in order

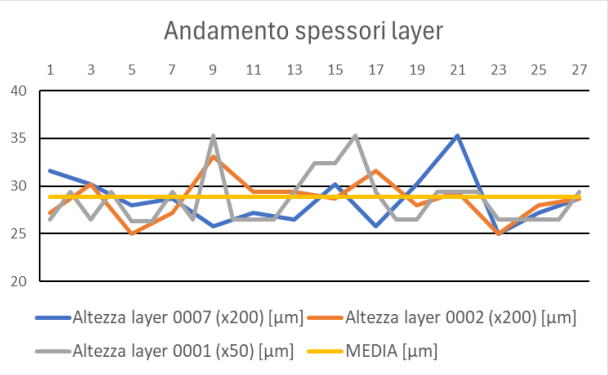
[3] Procedure for preparing specimens for the electron microscope: in order for the sample to be observed through the electron microscope, it must be able to conduct electrical energy; for this reason, the surface of the samples is metallized with a layer of gold, about 40 nm thick, by immersing it in Argon plasma inside a vacuum chamber.

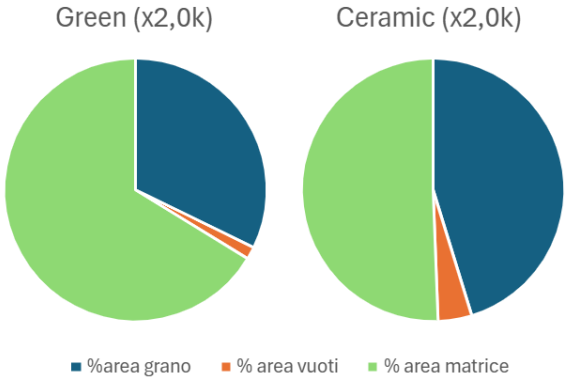
to obtain a ceramic piece. We measure the percentage shrinkage as the ratio between the difference of the two sizes and the starting size.

Flexural bars	*Length*	*Width*	*Height*	*Area*	*Volume*
(CAD-Green)/CAD	1,5%	-8,2%	12%	1,0%	6,2%
(Green-Ceramic)/Green	17,1%	15,5%	10,2%	28,4%	37,1%
(CAD-Ceramic)/CAD	18,3%	8,6%	21%	29,1%	41,0%





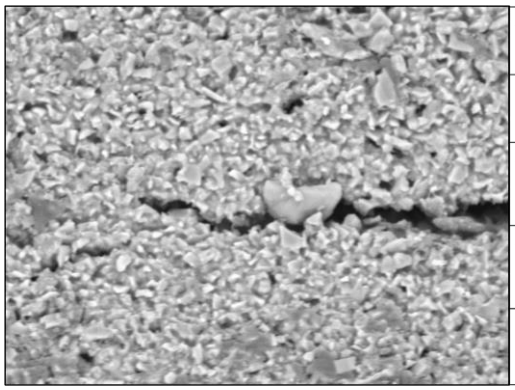




The height of the layers was measured from three images with different magnifications under an electron microscope using a specially written MATLAB® code . A statistical analysis, which also allowed the code to be appropriately calibrated, resulted in a height of 28.836 ± 0.332 µm, compared to a print height set at 30 µm. [4] By analyzing images at different magnifications, a statistical estimate of the size of the grains and voids is obtained. The grain axis goes from 0.9335 ± 0.0297 µm to 0.9767 ± 0.079 µm, while that of the voids goes from 0.5034 ± 0.0172 µm to 0.7615 ± 0.0269 µm: sintering has led to the transport of material and fusion between neighboring particles, enlarging both the grains and the voids. By calibrating the code appropriately, the percentage of grains, matrix and voids is estimated. There is a decrease in the percentage of matrix between green and ceramic sample, although remaining partly present in the ceramic sample as well. There is also a greater presence of voids in the ceramic sample compared to the green sample.

[4] The uncertainties reported in this section are calculated as the ratio between the sample standard deviation and the square root of the number of samples observed, since they are uncertainties deriving from a statistical analysis of data collected

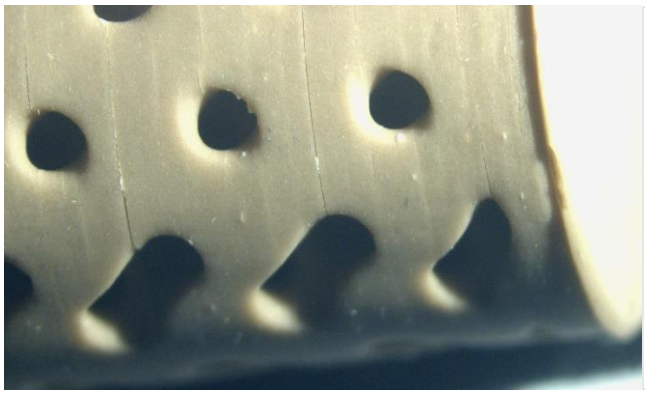
From the electron microscope analysis, cracks are observed in both the green and sintered samples. Larger particles are the cause of the propagation of cracks in the green sample, helped by the fact that the particles are not fused together. Cracks in the ceramic sample are caused by large regolith grains, which did not fuse with each other in sintering.

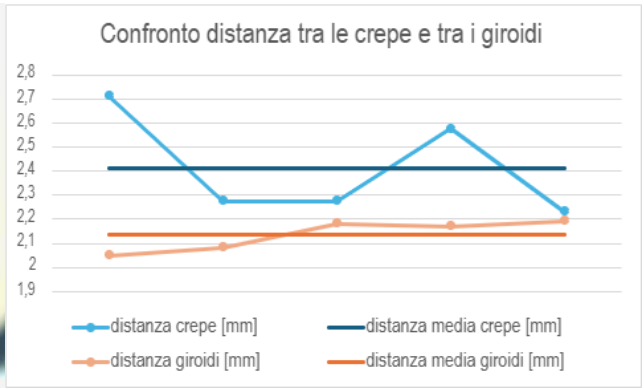


Area [µm²]	70,877
Perimetro [µm]	37,289
Asse maggiore [µm]	13,7530
Asse minore [µm]	7,0284
Eccentricità [-]	0,85991

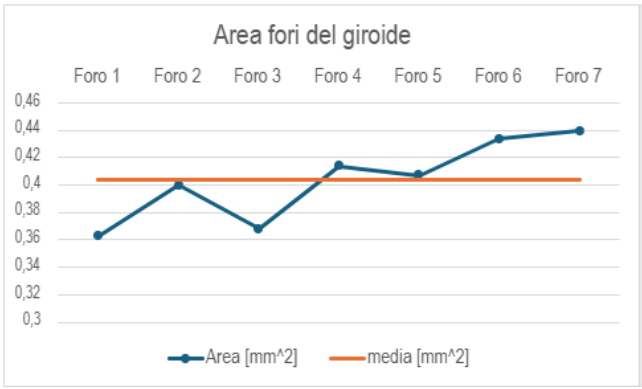
These cause the propagation of breaks, which are caused by the concentration of stresses during the sintering process. An example of a large grain size is presented in the figure. The grain has a major axis of 13.753 µm and is considered large compared to the maximum size of the measured regolith powder of about 8.710 µm. The crack generated is more than 1882 µm long (limit imposed by the size of the image itself). An average thickness of the cracks observed of 13.098 µm is also measured (the data does not have a measure of uncertainty, as the measurements collected did not allow a relevant statistical study).

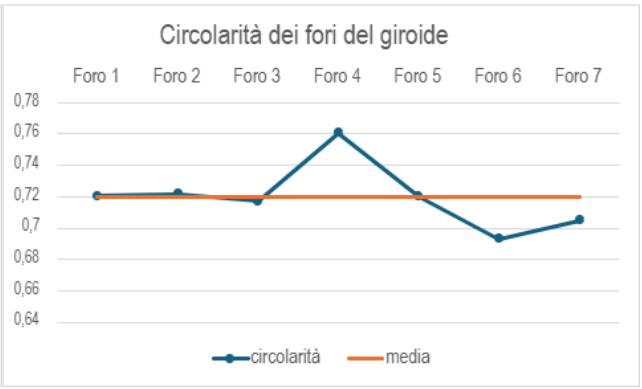
In the images of the sintered cylinders, cracks are observed at regular intervals, which can be compared with the distances between the holes due to gyroids. It is assumed that this is caused by the low percentage of material in the corresponding layer.

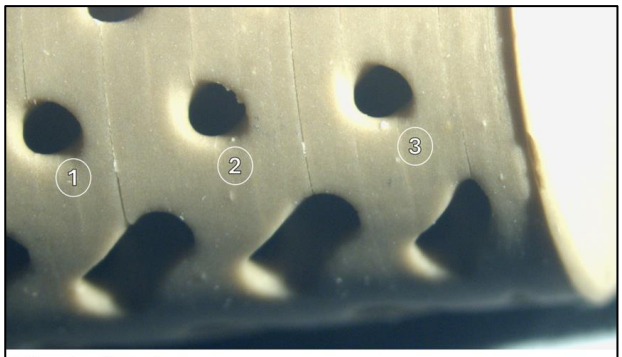




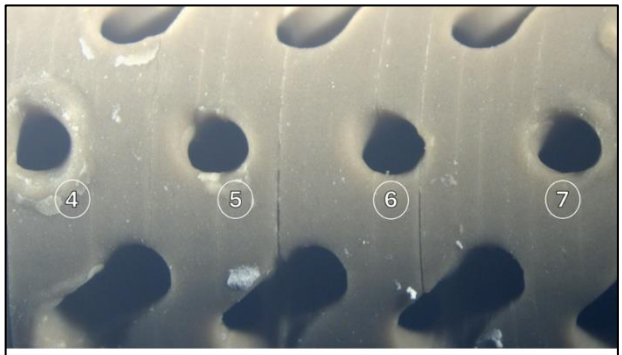
From the images under the optical microscope it is possible to measure the area and circularity of the holes due to the gyroids of the two sintered cylinders.







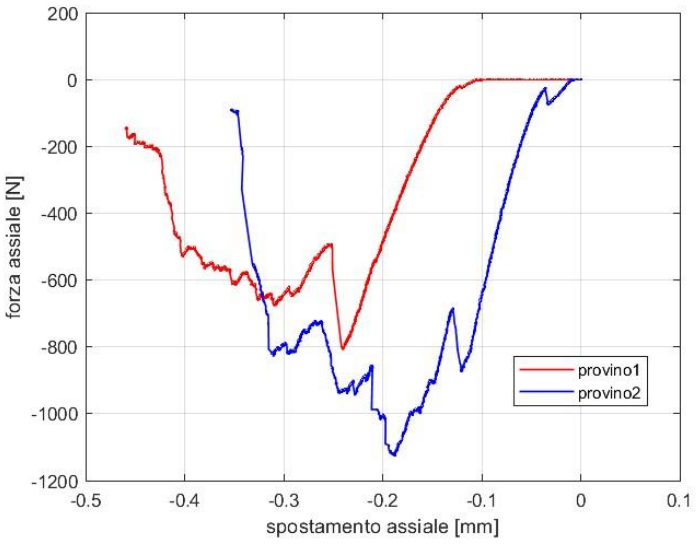
5A_sin_5_1.1x

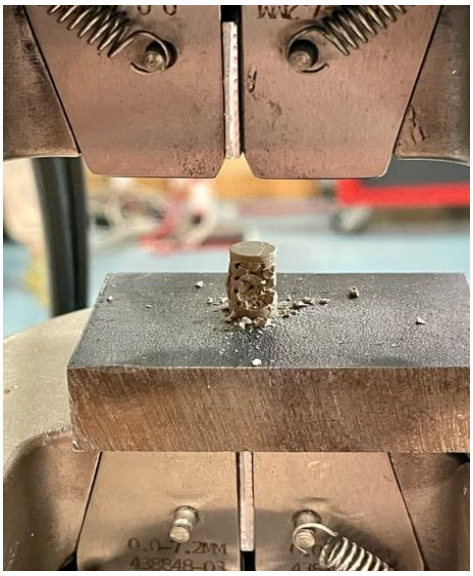


5B_ceram_1_1.1x

3.2. Mechanical Properties

To evaluate the mechanical properties of the components produced, compression and bending tests were conducted in the laboratory. Two samples of cylinders with gyroid lattice machining were tested at compression to failure with the aid of the *MTS 858 Mini Bionix II*® system, measuring the axial force impressed and the vertical displacement over time (shown below in the graph).

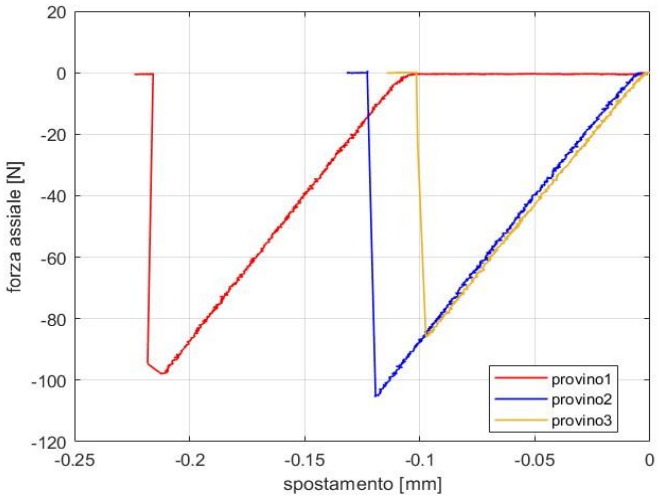




Subsequently, the data were used according to the indications of the C1424−25 standard for the calculation of the compressive strengths of the specimens which are worth 109.8726 MPa and 153.0380 Mpa respectively. Since the tested cylinders have a 60% "gyroid infill" machining, the resistance area considered is the minimum one, equal to 60% of the nominal one. In addition, the stiffness values in the linear sections were estimated: 7.2938 kN/mm and 10.967 kN/mm. The stress-strain curve has not been derived as no further data are available.

Three samples of bars were tested flexurally to failure with the "Three-Point Fixture" configuration described in the C1161−18 standard.

The data obtained were used for the calculation of the flexural strength with the standard formula indicated by the standard for the chosen configuration. The relative curves and the flexural strength values of the three tested specimens are shown below.



Flexural strength
Specimen 1
S1=134.4267 MPa
Specimen 2
S2=147.9392 MPa
Specimen 3
S3=114.9294 MPa

In both cases, it was not possible to conduct a statistical analysis of the mechanical properties for the limited number of samples available.

4. Discussion and conclusions

The quality analysis led to a detailed analysis of the cracks developed in the parts produced.
The study of defectology is carried out systematically through the Ishikawa diagram, from which the human component is excluded. The birth of cracks is linked to:

- Machine error
  Incomplete detachment of the print bed and any mechanical stress in the part can result from the choice of excessive *lift* and *retract* speeds.
- Method error
  Incomplete curing can be caused by insufficient exposure time and/or too short waiting times.
- Material error
  The likelihood of print defect agglomeration is increased by the small particle size.

The possible errors described contribute to the creation of less accurate components, highlighting the limitations deriving from the production process followed. In addition, the necessary manual definition of the code parameters for reading the images, although supported by a statistical analysis, constitutes a possible limitation in the acquisition of measurement data.

The presence of all these problems and random and non-eliminable errors makes the process not perfectly replicable, this is also demonstrated by the inconsistency between the bending failure points of the bars and the different compression behavior of the cylinders.
A mechanical testing campaign with a statistically significant number of specimens could provide more robust indications.

The subject of further study could be the study of how the printing orientation, the size of the layers and gyroids influence the mechanical properties and quality of the molded parts.

5. Bibliography

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4. Effect of particle size on rheology, curing kinetics, and corresponding mechanical and thermal properties of aluminum nitride (AlN) ceramic by digital light processing (DLP)-based vat photopolymerization Authors: Lifu Lin, Haidong Wu, Yehua Li, Junye Wang, Shanghua Wu, Journal of the European Ceramic Society, Volume 44, Issue 1, January 2024, Pages 184-192

5. C1161 − 18 (Reapproved 2023): Standard Test Method for Flexural Strength of Advanced Ceramics at Ambient Temperature

6. C1424 − 25: Standard Test Method for Monotonic Compressive Strength of Advanced Ceramics at Ambient Temperature

7. Image Processing Onramp® (MathWorks®)

8. Image Processing with MATLAB® (MathWorks®)